Chapter 2

Binary Response

2.1 Heart Disease Example

What might affect the chance of getting heart disease? One of the earliest studies addressing this issue started in 1960 and used 3154 healthy men, aged from 39 to 59, from the San Francisco area. At the start of the study, all were free of heart disease. Eight and a half years later, the study recorded whether these men now suffered from heart disease along with many other variables that might be related to the chance of developing this disease. We load a subset of this data from the Western Collaborative Group Study described in Rosenman et al. (1975):

```
data(wcgs, package="faraway")
```

We start by focusing on just three of the variables in the dataset:

```
summary(wcgs[,c("chd","height","cigs")])
```

```
  chd          height          cigs
 no :2897   Min.   :60.0   Min.   : 0.0
 yes: 257   1st Qu.:68.0   1st Qu.: 0.0
            Median :70.0   Median : 0.0
            Mean   :69.8   Mean   :11.6
            3rd Qu.:72.0   3rd Qu.:20.0
            Max.   :78.0   Max.   :99.0
```

We see that only 257 men developed heart disease as given by the factor variable `chd`. The men vary in `height` (in inches) and the number of cigarettes (`cigs`) smoked per day. We can plot these data using R base graphics:

```
plot(height ~ chd, wcgs)
wcgs$y <- ifelse(wcgs$chd == "no",0,1)
plot(jitter(y,0.1) ~ jitter(height), wcgs, xlab="Height",  ylab="Heart
    ↪  Disease", pch=".")
```

The first panel in Figure 2.1 shows a boxplot. This shows the similarity in the distribution of heights of the two groups of men with and without heart disease. But the heart disease is the response variable so we might prefer a plot which treats it as such. We convert the absence/presence of disease into a numerical 0/1 variable and plot this in the second panel of Figure 2.1. Because heights are reported as round numbers of inches and the response can only take two values, it is sensible to add a small amount of noise to each point, called *jittering*, so that we can distinguish them. Again we can see the similarity in the distributions. We might think about fitting a line to this plot.

More informative plots may be obtained using the `ggplot2` package of Wickham (2009). In the first panel of Figure 2.2, we see two histograms showing the distribution of heights for both those with and without heart disease. The *dodge* option ensures that the two histograms are interleaved. We see that the two distributions are

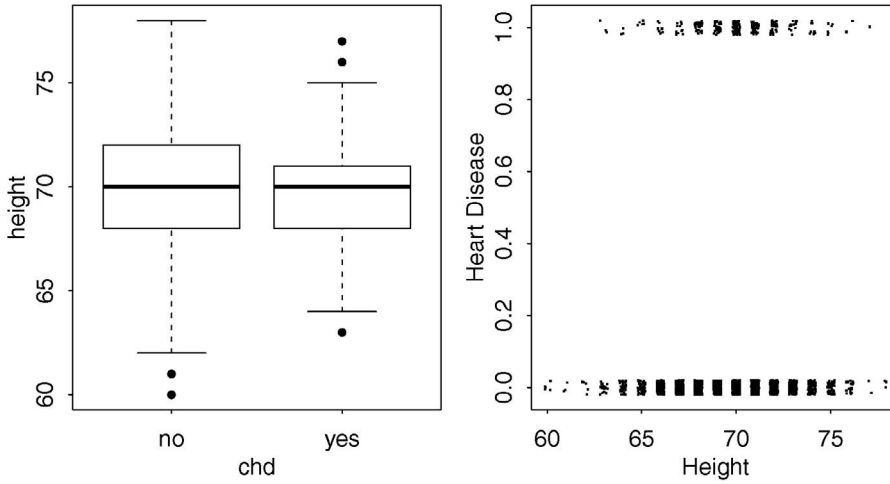


Figure 2.1 *Plots of the presence/absence of heart disease according to height in inches.*

similar. We also had to set the bin width of the histogram. It was natural to use one inch as all the height measurements are rounded to the nearest inch. In the second panel of Figure 2.2, we see the corresponding histograms for smoking. In this case, we have shown the frequency rather than the count version of the histogram. We see that smokers are more likely to get heart disease.

```
library(ggplot2)
ggplot(wcgs, aes(x=height, color=chd)) + geom_histogram(position="
    ↪ dodge", binwidth=1)
ggplot(wcgs, aes(x=cigs, color=chd)) + geom_histogram(position="dodge"
    ↪ , binwidth=5, aes(y=..density..))
```

It is also helpful to plot both predictors and the response on the same display as seen in Figure 2.3. Some special effort is necessary in constructing this plot because of the large number of men that have the same height and cigarette usage. We use both jittering and partial transparency, controlled by the parameter `alpha` to provide a clearer sense of the numbers of such repeated points. With a smaller number of points, it is possible to combine both levels of the response in a single plot by using two colors or plotting symbols. But this does not work well with a large number of points as it becomes difficult to distinguish the two. We have chosen to display the two levels on separate panels called *facets* in the `ggplot2` package. It is worth the effort of plotting data well to gain insight into the best approaches to modeling.

```
ggplot(wcgs, aes(x=height,y=cigs))+geom_point(alpha=0.2, position=
    ↪ position_jitter())+facet_grid(~ chd)
```

We would like to predict the heart disease outcome for a given individual and also to explain the relationship between height, cigarette usage and heart disease. We observe that, for the same height and cigarette consumption, both outcomes occur. This occurs quite regularly. Hence it makes better sense to model the probability of getting heart disease rather than the outcome itself.

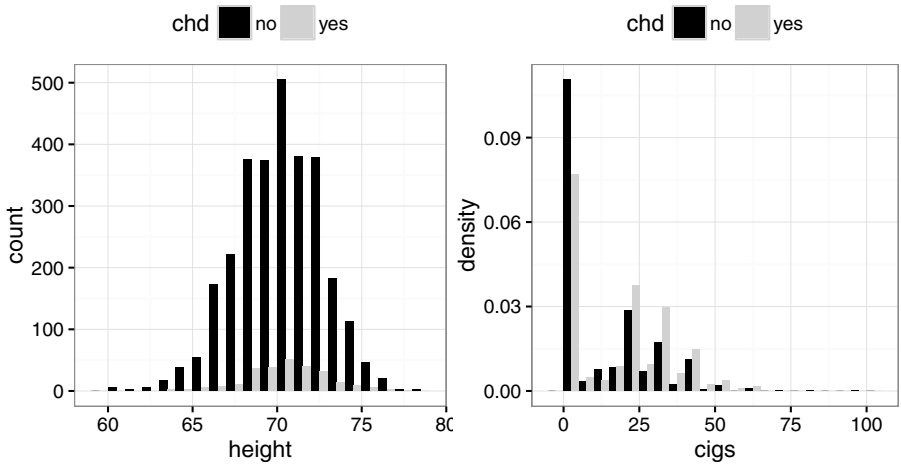


Figure 2.2 *Interleaved histograms of the distribution of heights and cigarette usage for men with and without heart disease.*

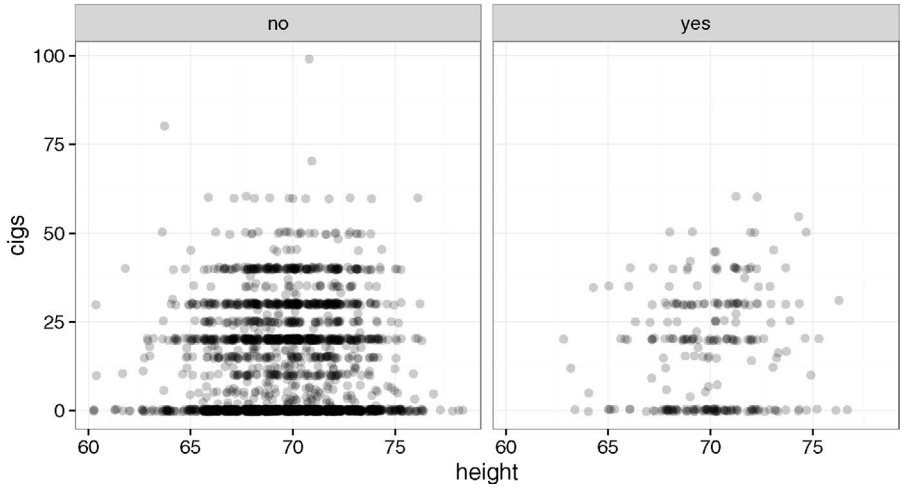


Figure 2.3 *Height and cigarette consumption for men without heart disease on the left and with heart disease on the right. Some jittering and transparency have been used to reduce overplotting problems.*

We might envisage fitting a line to the data in the second panel of Figure 2.1. But there are several problems with this approach. A line would eventually extend above one or below zero and these are not valid probabilites. We could truncate the predictions in these regions to $[0,1]$ but this would result in predictor values where the outcome was considered certain. That does not seem reasonable in this dataset, nor would it in many others. So using a linear model for binary response data is usually not a sensible idea.

2.2 Logistic Regression

Suppose we have a response variable Y_i for $i = 1,\ldots,n$ which takes the values zero or one with $P(Y_i = 1) = p_i$. This response may be related to a set of q predictors $(x_{i1},\ldots,x_{iq})$. We need a model that describes the relationship of $x_1,\ldots,x_q$ to the probability p. Following the linear model approach, we construct a *linear predictor*:

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_q x_{iq}$$

Since the linear predictor can accommodate quantitative and qualitative predictors with the use of dummy variables and also allows for transformations and combinations of the original predictors, it is very flexible and yet retains interpretability. The idea that we can express the effect of the predictors on the response solely through the linear predictor is important. The idea can be extended to models for other types of response and is one of the defining features of the wider class of generalized linear models (GLMs) discussed later in Chapter 8.

We have seen previously that the linear relation $\eta_i = p_i$ is not workable because we require $0 \le p_i \le 1$. Instead we shall use a *link function* g such that $\eta_i = g(p_i)$. We need g to be monotone and be such that $0 \le g^{-1}(\eta) \le 1$ for any η. The most popular choice of link function in this situation is the *logit*. It is defined so that:

$$\eta = \log(p/(1-p))$$

or equivalently:

$$p = \frac{e^\eta}{1+e^\eta}$$

Combining the use of the logit link with a linear predictor gives us the term *logistic regression*. Other choices of link function are possible but we will defer discussion of these until later. The logit and its inverse are defined as `logit` and `ilogit` in the `faraway` package. The relationship between p and the linear predictor η is shown in Figure 2.4.

```
library(faraway)
curve(ilogit(x),-6,6, xlab=expression(eta), ylab="p")
```

The logistic curve is almost linear in its midrange. This means that for modeling responses that are all of moderate probability, logistic and linear regression will not behave very differently. The curve approaches one at its upper end and zero at its lower end but never reaches these bounds. This means that logistic regression will never predict that anything is inevitable or impossible. Notice that changes in the linear predictor will result in different changes in the probability depending on where

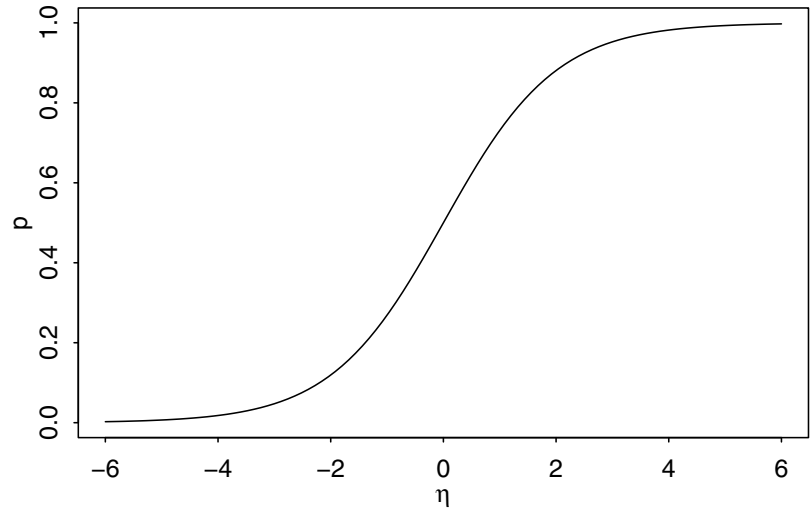


Figure 2.4 *A logistic relationship between the probability of the response, p, and the linear predictor,* η.

the change is made. This is in contrast to a linear relationship and makes the interpretation of logistic regression coefficients more difficult.

Now we estimate the parameters of the model. We use the method of maximum likelihood; see Appendix A for a brief introduction to this method. The log-likelihood using the logit link is given by:

$$l(\beta) = \sum_{i=1}^{n} \left[y_i \eta_i - \log(1 + e_i^{\eta}) \right]$$

We can maximize this to obtain the maximum likelihood estimates $\hat{\beta}$ and use the standard theory to obtain approximate standard errors. An algorithm to perform the maximization will be discussed in Chapter 8. We fit the model in R like this:

```
lmod <- glm(chd ~ height + cigs, family = binomial, wcgs)
```

This model is a special case of wider class called a generalized linear model (GLM) so we use the `glm` fitting function and specify the distribution of the response, in this case binomial, which identifies the member of this wider family of models. The model formula is specified in the same way as for linear models. A binary response is a special case of the binomial and requires that the response have only two levels. By default, the first level alphabetically will be associated with $y = 0$ and the second with $y = 1$. If you want to change this, use the `relevel` command.

We can examine the standard summary output:

```
summary(lmod)
Call:
glm(formula = chd ~ height + cigs, family = binomial, data = wcgs)

Deviance Residuals:
   Min      1Q  Median      3Q     Max
```

```
-1.004  -0.443  -0.363  -0.350   2.436

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.50161    1.84186   -2.44    0.015
height       0.02521    0.02633    0.96    0.338
cigs         0.02313    0.00404    5.72    1e-08

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1781.2  on 3153  degrees of freedom
Residual deviance: 1749.0  on 3151  degrees of freedom
AIC: 1755

Number of Fisher Scoring iterations: 5
```

A shorter version is available in the `faraway` package as `sumary` (a shortened summary). Incidentally, we don't need to keep loading the `faraway` package within a single R session. We have included it here just as a reminder of where the `sumary` function comes from. If you get a "not found" message, it means you forgot to load this package.

```
library(faraway)
sumary(lmod)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.50161    1.84186   -2.44    0.015
height       0.02521    0.02633    0.96    0.338
cigs         0.02313    0.00404    5.72    1e-08

n = 3154 p = 3
Deviance = 1749.049 Null Deviance = 1781.244 (Difference = 32.195)
```

You can always use the `summary` version if you prefer but henceforth we will use this shorter `sumary`. Let's start with the interpretation of the regression coefficients: $\hat{\beta}_0 = -4.50$, $\beta_1 = 0.025$ and $\beta_2 = 0.023$. We can compute the probability of heart disease given the values of the predictors. We can vary the height for fixed levels of cigarette consumption — nonsmoker and 20 a day. For this we need to extract the coefficients:

```
(beta <- coef(lmod))
```

```
(Intercept)      height        cigs
  -4.501614    0.025208    0.023127
```

We then construct the plot using a point plotting symbol and add the fitted curves:

```
plot(jitter(y,0.1) ~ jitter(height), wcgs, xlab="Height", ylab="Heart
    ↪ Disease",pch=".")
curve(ilogit(beta[1] + beta[2]*x + beta[3]*0),add=TRUE)
curve(ilogit(beta[1] + beta[2]*x + beta[3]*20),add=TRUE,lty=2)
```

In the first panel of Figure 2.5, we see that the probability of heart disease increases slightly with height. We also see a somewhat higher risk for smokers. We produce a similar plot varying the cigarette consumption rather than the height:

```
plot(jitter(y,0.1) ~ jitter(cigs), wcgs, xlab="Cigarette Use", ylab="
    ↪ Heart Disease",pch=".")
curve(ilogit(beta[1] + beta[2]*60 + beta[3]*x),add=TRUE)
curve(ilogit(beta[1] + beta[2]*78 + beta[3]*x),add=TRUE,lty=2)
```

In the second panel of Figure 2.5, we see how the probability of heart disease increases with the amount of smoking.

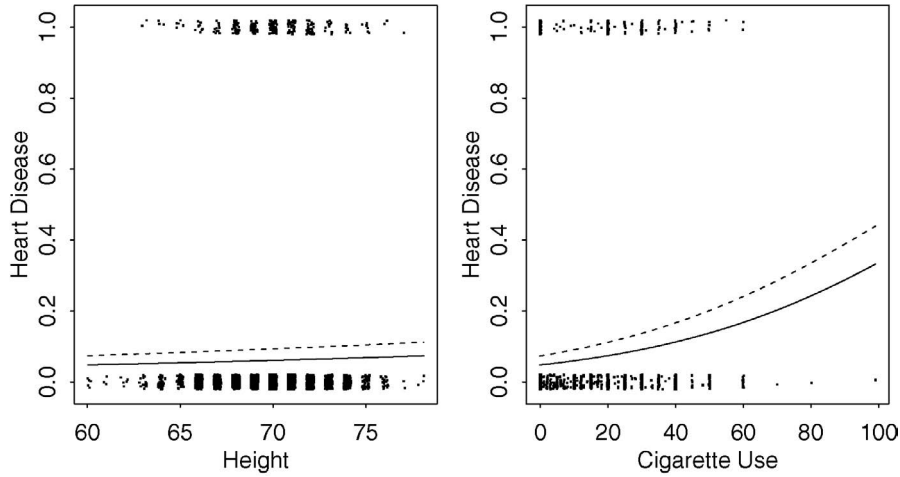


Figure 2.5 *Predicted probability of heart disease as height and cigarette consumption vary. In the first panel, the solid line represents a nonsmoker, while the dashed line is a pack-a-day smoker. In the second panel, the solid line represents a very short man (60 in. tall) while the dashed line represents a very tall man (78 in. tall.)*

Interpreting Odds: Odds are an alternative scale to probability for representing chance. They arose as a way to express the payoffs for bets. An *evens* bet means that the winner gets paid an equal amount to that staked. A 3–1 *against* bet would pay \$3 for every \$1 bet, while a 3–1 *on* bet would pay only \$1 for every \$3 bet. If these bets are *fair* in the sense that a bettor would break even in the long-run average, then we can make a correspondence to probability. Let p be the probability and o be the odds, where we represent 3–1 against as 1/3 and 3–1 on as 3, then the following relationship holds:

$$\frac{p}{1-p} = o \qquad \texttt{or} \qquad p = \frac{o}{1+o}$$

One mathematical advantage of odds is that they are unbounded above, which makes them more convenient for some modeling purposes.

Odds also form the basis of a subjective assessment of probability. Some probabilities are determined from considerations of symmetry or long-term frequencies, but such information is often unavailable. Individuals may determine their subjective probability for events by considering what odds they would be prepared to offer on the outcome. Under this theory, other potential persons would be allowed to place bets for or against the event occurring. Thus the individual would be forced to make an honest assessment of probability to avoid financial loss.

If we have two covariates x_1 and x_2, then the logistic regression model is:

$$\log(\text{odds}) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

or

$$\text{odds} = e^{\beta_0}.e^{\beta_1 x_1}.e^{\beta_2 x_2}$$

Now β_1 can be interpreted as follows: a unit increase in x_1 with x_2 held fixed increases the log-odds of success by β_1 or increases the odds of success by a factor of e^{β_1}. So the exponentiated coefficients are more useful:

```
exp(beta)
```
```
(Intercept)      height        cigs
   0.011091    1.025528    1.023397
```

We can say that the odds of heart disease increase by 2.6% with each additional inch in height and by 2.3% with each additional cigarette smoked per day. Note that $\exp x \approx 1 + x$ for small values of x. We observe that $\hat{\beta}_2 = 0.023$ so the 2.3% increase in odds due to smoking a cigarette a day could have been quickly estimated from the original model output without further computation. This approximation is useful for quick intuitions.

It is more natural to compute the effect of a pack a day (20 cigarettes):

```
exp(beta[3]*20)
```
```
  cigs
1.5881
```

So we have 59% increase in the odds of heart disease due to smoking. Of course, the usual interpretational difficulties regarding causation apply as in standard regression. We cannot conclude, just from this study, that smoking causes an increase in the risk of heart disease.

Most people are more comfortable with a probability scale so it is a good idea to compute the difference in the predicted probabilities as a predictor is changed over a sensible range as we have done above. Odds are more convenient from a mathematical perspective and do allow us to express the effects of smoking in this example in a compact way. One disadvantage of odds ratios is that they only express the relative difference. Although smoking is associated with a strongly increased risk of heart disease, the absolute probabilites are still not large.

An alternative notion to odds is relative risk. Suppose the probability of "success" in the presence of some condition is p_1 and p_2 in its absence. The relative risk is p_1/p_2. For example, the predicted probabilities of getting heart disease for a 68in.-tall man who does not smoke and who smokes 20 a day are, respectively:

```
c(ilogit(sum(beta*c(1,68,20))),ilogit(sum(beta*c(1,68,0))))
```
```
[1] 0.089079 0.058004
```

The relative risk is then:

```
ilogit(sum(beta*c(1,68,20)))/ilogit(sum(beta*c(1,68,0)))
```
```
[1] 1.5357
```

The relative risk of 1.54 is not far different from the odds ratio of 1.59. For low probability outcomes, the relative risk and the odds ratio will be very similar, but for larger probabilities, there may be substantial differences.

2.3 Inference

Consider two models, a larger model with l parameters and likelihood L_L and a smaller model with s parameters and likelihood L_S where the smaller model represents a subset (or more generally a linear subspace) of the larger model. Likelihood

methods suggest the likelihood ratio statistic:

$$2\log\frac{L_L}{L_S} \tag{2.1}$$

as an appropriate test statistic for comparing the two models. Now suppose we choose a saturated larger model — such a model typically has as many parameters as cases and has fitted values $\hat{p}_i = y_i$. The test statistic becomes:

$$D = -2\sum_{i=1}^{n} \hat{p}_i \text{logit}(\hat{p}_i) + \log(1-\hat{p}_i)$$

where $\hat{p}_i$ are the fitted values from the smaller model. D is called the *deviance* and is useful in making hypothesis tests to compare models.

In other examples of GLMs, the deviance is a measure of how well the model fit the data but in this case, D is just a function of the fitted values $\hat{p}$ so it cannot be used for that purpose. Other methods must be used to judge goodness of fit for binary data — for example, the Hosmer-Lemeshow test described in Section 2.6.

In the summary output previously, we had:

```
Deviance = 1749.049 Null Deviance = 1781.244 (Difference = 32.195)
```

The `Deviance` is the deviance for the current model while the `Null Deviance` is the deviance for a model with no predictors and just an intercept term.

We can use the deviance to compare two nested models. The test statistic in (2.1) becomes $D_S - D_L$. This test statistic is asymptotically distributed χ^2_{l-s}, assuming that the smaller model is correct and the distributional assumptions hold. For example, we can compare the fitted model to the null model (which has no predictors) by considering the difference between the residual and null deviances. For the heart disease example, this difference is 32.2 on two degrees of freedom (one for each predictor). Hence, the p-value for the test of the hypothesis that at least one of the predictors is related to the response is:

```
1-pchisq(32.2,2)
[1] 1.0183e-07
```

Since this value is so small, we are confident that there is some relationship between the predictors and the response. Note that the expected value of a χ^2-variate with d degrees of freedom is simply d so we knew the p-value would be small before even calculating it.

We can test the individual predictors by fitting models that drop these predictors and computing the difference in the deviance observed. We test the significance of height in the model as follows:

```
lmodc <- glm(chd ~ cigs, family = binomial, wcgs)
anova(lmodc,lmod, test="Chi")
Analysis of Deviance Table

Model 1: chd ~ cigs
Model 2: chd ~ height + cigs
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      3152       1750
2      3151       1749  1     0.92     0.34
```

The analysis of deviance table displays all the information. We see that height is not significant in a model that already includes cigarette consumption. We can readily test all the predictors in the model using the `drop1` function:

```
drop1(lmod,test="Chi")
Single term deletions

Model:
chd ~ height + cigs
       Df Deviance  AIC   LRT Pr(>Chi)
<none>        1749 1755
height  1     1750 1754  0.92     0.34
cigs    1     1780 1784 31.07  2.5e-08
```

An alternative to this test is the z-value, which is $\hat{\beta}/se(\hat{\beta})$, which is approximately normally distributed. Recalling the output from before:

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.50161    1.84186   -2.44    0.015
height       0.02521    0.02633    0.96    0.338
cigs         0.02313    0.00404    5.72    1e-08
```

In this case, the outcomes are similar to previous tests but not identical. This is in contrast to the normal (Gaussian) linear model where the two would be identical. In this particular example, there is no practical difference, but in some cases, especially with sparse data, the standard errors can be overestimated and so the z-value is too small and the significance of an effect could be missed. This is known as the Hauck–Donner effect — see Hauck and Donner (1977). So the deviance-based test is preferred.

Confidence intervals for the regression parameters can be constructed using normal approximations for the parameter estimates. A $100(1-\alpha)\%$ confidence interval for β_i would be:

$$\hat{\beta}_i \pm z^{\alpha/2} se(\hat{\beta}_i)$$

where $z^{\alpha/2}$ is a quantile from the normal distribution. Thus a 95% confidence interval for β_1 in our model would be:

```
0.02521 + c(-1,1) * 1.96 * 0.02633
[1] -0.026397  0.076817
```

A better way is to construct a profile likelihood-based confidence interval:

```
confint(lmod)
Waiting for profiling to be done...
                2.5 %    97.5 %
(Intercept) -8.134755 -0.912970
height      -0.026199  0.077028
cigs         0.015149  0.031005
```

The profile likelihood method is generally preferable for the same Hauck–Donner reasons discussed above although it is more work to compute.

2.4 Diagnostics

Regression diagnostics are useful in checking the assumptions of the model and in identifying any unusual points. As with linear models, residuals are the most important means of determining how well the data fits the model and where any changes or

improvements might be advisable. We can compute residuals as a difference between observed and fitted values. There are two kinds of fitted (or predicted) values:

```
linpred <- predict(lmod)
predprob <- predict(lmod, type="response")
```

The former is the predicted value in the linear predictor scale, η, while the latter is the predicted probability $p = \text{logit}^{-1}(\eta)$. Here are the first few values and a confirmation of the relationship between them:

```
head(linpred)
```

```
   2001    2002    2003    2004    2005    2006
-2.0833 -2.2745 -2.7623 -2.3249 -2.2745 -2.6867
```

```
head(predprob)
```

```
    2001     2002     2003     2004     2005     2006
0.110734 0.093255 0.059397 0.089079 0.093255 0.063766
```

```
head(ilogit(linpred))
```

```
    2001     2002     2003     2004     2005     2006
0.110734 0.093255 0.059397 0.089079 0.093255 0.063766
```

We compute the *raw residuals* as $y - \hat{p}$:

```
rawres <- wcgs$y - predprob
```

These can also be obtained as `residuals(lmod, type="response")`. Following the standard practice for diagnostics in linear models, we plot the residuals against the fitted values:

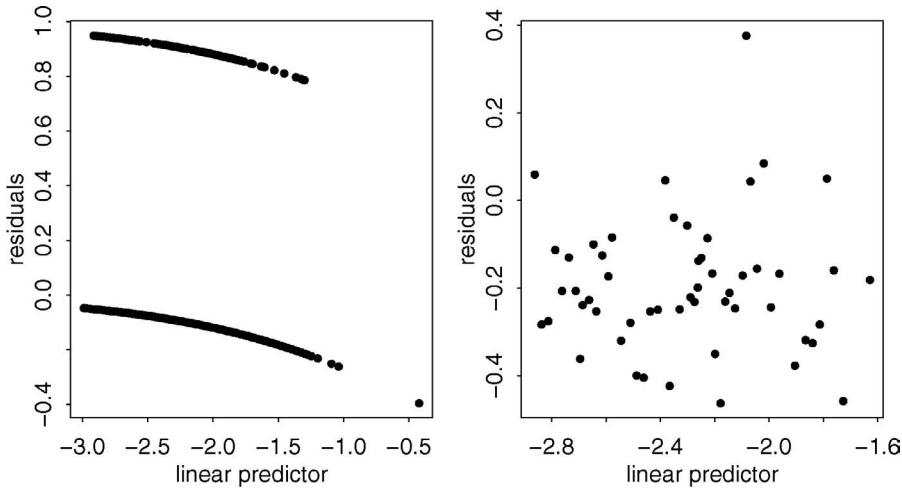


Figure 2.6 *The panel on the left shows the raw residuals and linear predictor. The two lines are due to the binary response. The panel on the right shows the binned version of the plot.*

```
plot(rawres ~ linpred, xlab="linear predictor", ylab="residuals")
```

The plot, as seen in the first panel of Figure 2.6, is not very helpful. Because $y = 0$ or 1, the residual can take only two values given a fixed linear predictor. The upper line in the plot corresponds to $y = 1$ and the lower line to $y = 0$. We gain no insight into the fit of the model. We have chosen to plot the linear predictor rather than the predicted probability on the horizontal axis because the former provides a better spacing of the points in this direction.

Another problem with the raw residuals is that we do not expect them to have equal variance. Binary variance is given by $p(1-p)$ so there will be more variation from probabilities in the midrange. Some form of standardization is desirable. In the standard linear model, the residual sum of squares is given by $\sum_i \hat{\varepsilon}_i^2$. For a logistic regression model, the equivalent quantity is the deviance. The *deviance residuals*, r_i, are defined by analogy using $\sum r_i^2 =$ Deviance. We ensure these residuals have the right sign by defining:

$$r_i = sign(y_i - \hat{p}_i)\sqrt{r_i}^2$$

The deviance residuals are the default choice and are given by `residuals(lmod)`.

We now construct a more useful residuals plot by grouping the residuals into bins where the bins are based on similar predictor values. The choice of the number of bins depends on the size of the dataset. We choose 100 bins so that we have roughly 30 observations per bin. Some effort is required to construct this plot. The `dplyr` package of Wickham and Francois (2015) is useful for this task. First, we add the residuals and linear predictor into the data frame.

```
library(dplyr)
wcgs <- mutate(wcgs, residuals=residuals(lmod), linpred=predict(lmod))
```

The next step is to create the bins:

```
gdf <- group_by(wcgs, cut(linpred, breaks=unique(quantile(linpred,
    ↪ (1:100)/101))))
```

The `cut` function divides a variable into a factor with levels defined by breakpoints given by `breaks`. We would like to have a roughly equal number of points per bin so we divide the range of the linear predictor based on quantiles. As it happens, there are many tied values in this variable so we insist on only the `unique` breakpoints.

Now within each of these bins, we ask for the means of the residuals and linear predictors:

```
diagdf <- summarise(gdf, residuals=mean(residuals), linpred=mean(
    ↪ linpred))
```

The residuals plot can be seen in the second panel of Figure 2.6 as:

```
plot(residuals ~ linpred, diagdf, xlab="linear predictor")
```

The deviance residuals are not constrained to have mean zero so the mean level in the plot is not of interest. We see an even variation as the linear predictor varies so this plot reveals no inadequacy in the model. A similar plot can be constructed using the `binnedplot` function of the `arm` package of Gelman and Hill (2006).

We should also plot the binned residuals against the predictors. We make a similar calculation:

```
gdf <- group_by(wcgs, height)
diagdf <- summarise(gdf, residuals=mean(residuals))
ggplot(diagdf, aes(x=height,y=residuals)) + geom_point()
```

The height predictor only takes a limited number of integer values so we just group by height directly. The plot is shown in the first panel of Figure 2.7 where we see nothing remarkable except perhaps for a large residual for a height of 77 in. Let's look at the data for this height:

```
filter(wcgs, height==77) %>% select(height, cigs, chd, residuals)
```

```
  height cigs chd residuals
1     77    0  no  -0.38579
2     77    0 yes   2.29566
3     77    5  no  -0.40785
```

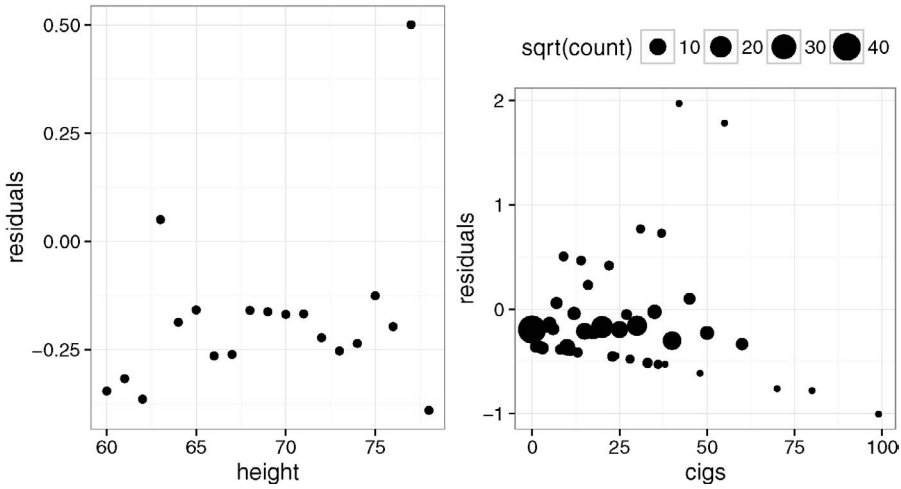


Figure 2.7 *Binned residuals plots for the predictors.*

We use some features from the `dplyr` package. The `filter` command picks out the subset of the rows of the data with height of 77 in. The output is piped to the next command using the operator `%>%`. The `select` command picks out the variables we want to see. From this we see that this bin consists of only three cases, of which one is a man who has heart disease. Hence, this point on the residual plot is not exceptional.

We now construct the corresponding plot for cigarette consumption as seen in the second panel of Figure 2.7.

```
group_by(wcgs, cigs) %>% summarise(residuals=mean(residuals), count=n
    ↪ ()) %>% ggplot(aes(x=cigs, y=residuals, size=sqrt(count))) +
    ↪ geom_point()
```

We use the `%>%` operator to chain together the intermediate steps. This time we keep a count of the number of residuals in each bin using the `n()` function within `summarise`. We use this count to control the size of the plotted point. We take square roots because the SD is proportional to the square root of the sample size so this gives the appropriate visual impression. Many men in the sample do not smoke at all so there is a large bin for zero consumption. Other round numbers such as 20 a day are similarly highlighted. We see the more unusual points on the display corresponding to small bin sizes and so they can be discounted in interpreting the plot. Focusing on the larger bins, we see no evidence of lack of fit or a need to transform the predictor.

We now turn attention to the detection of unusual points. A QQ plot of the residuals is standard practice in checking linear models. Such a plot is shown in the first panel of Figure 2.8.

```
qqnorm(residuals(lmod))
```

We see that the plot is very far from the desired linear relationship. We see two clusters of points corresponding to $y = 0$ and $y = 1$. But there is no reason to expect these residuals to be normally distributed so this does not raise any concern. The

largest residuals will be observed in cases where $y_i = 1$ when $\hat{p}_i$ is small and where $y_i = 0$ when $\hat{p}_i$ is close to 1. These represent occasions when something unexpected happened. But even in such cases, the residual cannot become particularly large.

We can detect cases which are unusual in the predictor space by examining the leverages just as in standard linear models. These are best examined using a half-normal plot as seen in the second panel of Figure 2.8.

```
halfnorm(hatvalues(lmod))
```

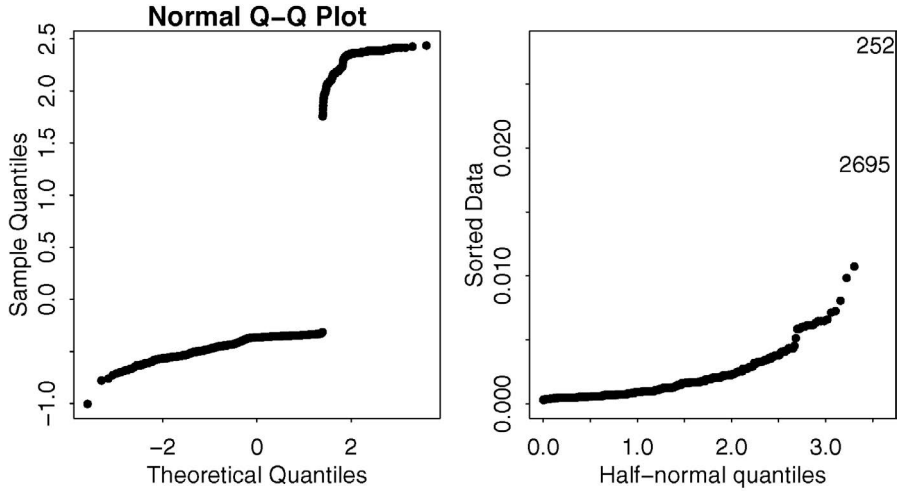


Figure 2.8 *A QQ plot of the deviance residuals is shown on the left and a half-normal plot of the leverages is shown on the right.*

The two outlying points can be identified:

```
filter(wcgs, hatvalues(lmod) > 0.015) %>% select(height, cigs, chd)
  height cigs chd
1     71   99  no
2     64   80  no
```

These are the two men with the very highest cigarette consumption which can be seen near the top of Figure 2.3. Given the relatively large size of the dataset and the fact that these two points are not particularly extreme, we are not concerned.

2.5 Model Selection

The analysis thus far has used only two of the predictors available but we might construct a better model for the response if we used some of the other predictors. We might find that not all these predictors are helpful in explaining the response. We would like to identify a subset of the predictors that model the response well without including any superfluous predictors.

We could use the inferential methods to construct hypothesis tests to compare various candidate models and use this as a mechanism for choosing a model. Back-

ward elimination is one such method which is relatively easy to implement. The method proceeds sequentially:

1. Start with the full model including all the available predictors. We can add derived predictors formed from transformations or interactions between two or more predictors.
2. Compare this model with all the models consisting of one less predictor. Compute the p-value corresponding to each dropped predictor. The `drop1` function in R can be used for this purpose.
3. Eliminate the term with largest p-value that is greater than some preset critical value, say 0.05. Return to the previous step. If no such term meets this criterion, stop and use the current model.

Thus predictors are sequentially eliminated until a final model is settled upon. Unfortunately, this is an inferior procedure. Although the algorithm is simple to use, it is hard to identify the problem to which it provides a solution. It does not identify the best set of predictors for predicting future responses. It is not a reliable indication of which predictors are the best explanation for the response. Even if one believes the fiction that there is a true model, this procedure would not be best for identifying such a model.

The Akaike information criterion (AIC) is a popular way of choosing a model — see Section A.3 for more. The criterion for a model with likelihood L and number of parameters q is defined by

$$AIC = -2\log L + 2q$$

We select the model with the smallest value of AIC among those under consideration. Any constant terms in the definition of log-likelihood can be ignored when comparing different models that will have the same constants. For this reason we can use AIC $=$ deviance $+ 2q$.

We would like to consider models consisting of all possible subsets of the available predictors. For large values of q, there may be a very large number of such subsets so a sequential search through this set may be necessary to reduce the computation. Such a sequential search is provided by the `step` command:

```
wcgs$bmi <- with(wcgs, 703*wcgs$weight/(wcgs$height^2))
lmod <- glm(chd ~ age + height + weight +bmi + sdp + dbp + chol +
    ↪ dibep + cigs +arcus, family=binomial, wcgs)
lmodr <- step(lmod, trace=0)
sumary(lmodr)
```

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -15.95760    2.28608   -6.98  2.9e-12
age            0.06159    0.01240    4.97  6.8e-07
height         0.05016    0.02782    1.80    0.071
bmi            0.06038    0.02660    2.27    0.023
sdp            0.01773    0.00415    4.27  2.0e-05
chol           0.01071    0.00153    7.01  2.4e-12
dibepB         0.65762    0.14590    4.51  6.6e-06
cigs           0.02104    0.00426    4.94  8.0e-07
arcuspresent   0.21100    0.14372    1.47    0.142

n = 3140 p = 9
Deviance = 1569.325 Null Deviance = 1769.171 (Difference = 199.846)
```

We have derived a body mass index (BMI) variable from height and weight. The factor of 703 is required to convert to the metric units used by BMI. The `trace=0` argument prevents the printing of the intermediate stages of the search. Omit this if you would like to see these. In this case, only two predictors are dropped from the full set: weight and diastolic blood pressure.

The urge to houseclean may tempt us to use model selection methods such as AIC. If the purpose of the model is to predict future observations, then AIC-based model selection may provide a decent choice for this purpose. There are other criteria, similar to AIC, that can also be used. On the other hand, we may be more interested in explaining the response. For example, we may wish to know what the risk factors are for coronary heart disease. We might be tempted to think that since diastolic blood pressure has been dropped from the model that is has no relation to the chance of heart disease. Even so, considered alone:

```
drop1(glm(chd ~ dbp, family=binomial, wcgs), test="Chi")
Single term deletions

Model:
chd ~ dbp
       Df Deviance  AIC  LRT Pr(>Chi)
<none>        1752 1756
dbp     1     1781 1783 29.6  5.5e-08
```

We see that diastolic blood pressure is related to the chance of heart disease. For this and other reasons, model selection is not a magical sword that can cut through the knotty problem of determining the risk factors for heart disease let alone what might be causing it.

2.6 Goodness of Fit

As mentioned earlier, we cannot use the deviance for a binary response GLM as a measure of fit. We can use diagnostic plots of the binned residuals to help us identify inadequacies in the model but these cannot tell us whether the model fits or not. Even so the process of binning can help us develop a test for this purpose. We divide the observations up into J bins based on the linear predictor. Let the mean response in the jth bin be y_j and the mean predicted probability be $\hat{p}_j$ with m_j observations within the bin. We compute these values:

```
wcgsm <- na.omit(wcgs)
wcgsm <- mutate(wcgsm, predprob=predict(lmod,type="response"))
gdf <- group_by(wcgsm, cut(linpred, breaks=unique(quantile(linpred,
    ↪ (1:100)/101))))
hldf <- summarise(gdf, y=sum(y), ppred=mean(predprob), count=n())
```

There are a few missing values in the data. The default method is to ignore these cases. The `na.omit` command drops these cases from the data frame for the purposes of this calculation. We use the same method of binning the data as for the residuals but now we need to compute the number of observed cases of heart disease and total observations within each bin. We also need the mean predicted probability within each bin.

When we make a prediction with probability p, we would hope that the event occurs in practice with that proportion. We can check that by plotting the observed

proportions against the predicted probabilities as seen in Figure 2.9. For a well-calibrated prediction model, the observed proportions and predicted probabilities should be close.

```
hldf <- mutate(hldf, se.fit=sqrt(ppred*(1-ppred)/count))
ggplot(hldf,aes(x=ppred,y=y/count,ymin=y/count-2*se.fit,ymax=y/count+2
    ↪ *se.fit))+geom_point()+geom_linerange(color=grey(0.75))+geom_
    ↪ abline(intercept=0,slope=1)+xlab("Predicted Probability")+ylab(
    ↪ "Observed Proportion")
```

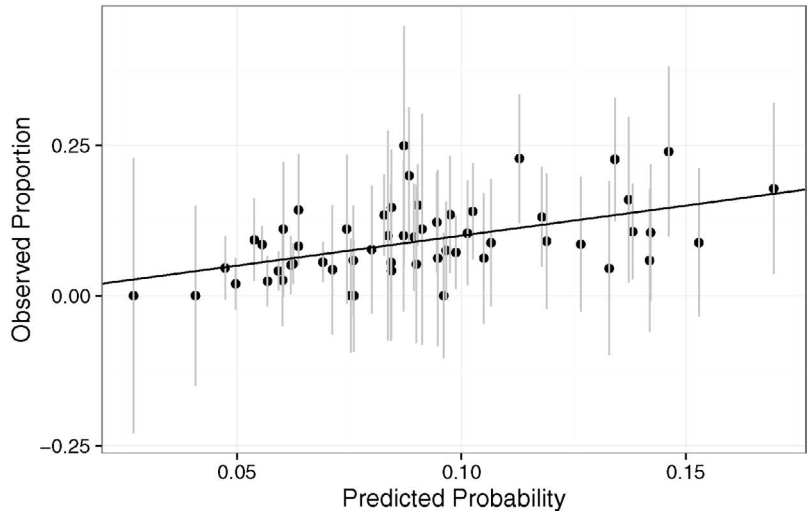


Figure 2.9 *Binned predicted probabilities and observed proportions for the heart disease model.*

Although we can see there is some variation, there is no consistent deviation from what is expected. We have computed approximate 95% confidence intervals using the binomial variation. The line passes through most of these intervals confirming that the variation from the expected is not excessive.

The Hosmer-Lemeshow statistic formalizes this assessment and is defined as:

$$X^2_{HL} = \sum_{j=1}^{J} \frac{(y_j - m_j\hat{p}_j)^2}{m_j\hat{p}_j(1-\hat{p}_j)}$$

This statistic has an approximate χ^2 distribution with $J-1$ degrees of freedom. We have some freedom to decide on the binning. We need sufficient observations per bin to ensure the accuracy of the χ^2 approximation yet not so few bins that the fit is barely tested.

```
hlstat <- with(hldf, sum( (y-count*ppred)^2/(count*ppred*(1-ppred))))
c(hlstat, nrow(hldf))
```

```
[1] 63.212 56.000
```

The p-value is then given by

```
1-pchisq(63.212, 56-1)
```

```
[1] 0.20898
```

Since the p-value is moderate, we detect no lack of fit. In marginal cases, it is worth experimenting with differing numbers of bins to check the robustness of the conclusion.

We might want a method for assessing the quality of predictions that does not depend on binning methods, as these involve some arbitrary choices about numbers and spacing of bins. *Scoring* methods were developed, initially by weather forecasters, to assess predictions. The most popular method is logarithmic scoring which is given as $y*log(\hat{p})+(1-y)*log(1-\hat{p})$. So if the event $y=1$ occurs, we score $\log\hat{p}$ and score $\log(1-\hat{p})$ if it does not. Better prediction methods have higher scores. Notice if we predict something is certain to happen ($\hat{p}=1$) but it does not occur then we score $-\infty$, so this method of scoring does not encourage certainty.

The model can be used to predict the outcome for each man in the dataset. Sometimes it is natural to think of the model as classifying observations. When $\hat{p}_i < 0.5$ we classify the case into no heart disease but when $\hat{p}_i \geq 0.5$ we put the case into the yes category. Of course, for the observed cases we already know the outcome but for future unknown cases, this classification may be helpful.

```
wcgsm <- mutate(wcgsm, predout=ifelse(predprob < 0.5, "no", "yes"))
xtabs( ~ chd + predout, wcgsm)
```

```
     predout
chd     no  yes
  no  2882    3
  yes  253    2
```

We see how the classifications match with the observed outcomes. The correct classification rate is:

```
(2882+2)/(2882+3+253+2)
```

```
[1] 0.91847
```

So the error or misclassification rate is about 8%. This can be a helpful way of judging the fit of the model although it can conceal some important variation. Suppose we were to use this model prospectively to predict the outcomes of men with given characteristics. Perhaps we might recommend certain treatments based on these predictions. For the men who will not develop heart disease, what fraction are predicted to not develop the disease? This is known as the *specificity* of the test. We calculate this as $2882/(2882+3)=0.999$, which is very high. In contrast, the proportion of those who will develop heart disease that are correctly identified by the model is called the *sensitivity*. Here this is $2/(253+2)=0.00784$, which is extremely low. Our prediction process is very unlikely to detect which men will develop heart disease. We see that the overall error rate of 8% conceals important information if this procedure were to be used as a diagnostic test.

We can improve the sensitivity of the test by dropping the 0.5 threshold we used for the classification. We compute how the sensitivity and specificity change as the threshold changes for a range of values up to 0.5:

```
thresh <- seq(0.01,0.5,0.01)
Sensitivity <- numeric(length(thresh))
Specificity <- numeric(length(thresh))
for(j in seq(along=thresh)){
    pp <- ifelse(wcgsm$predprob < thresh[j],"no","yes")
    xx <- xtabs( ~ chd + pp, wcgsm)
    Specificity[j] <- xx[1,1]/(xx[1,1]+xx[1,2])
```

```
    Sensitivity[j] <- xx[2,2]/(xx[2,1]+xx[2,2])
}
```

We can now plot how these change as a function of the threshold:

```
matplot(thresh,cbind(Sensitivity,Specificity),type="l",xlab="Threshold
    ↪ ",ylab="Proportion",lty=1:2)
```

The plot, shown in the first panel of Figure 2.10, demonstrates how the sensitivity falls but the specificity rises as the threshold is increased. The costs of the two possible kinds of errors in such diagnostic tests are rarely equal. We would need to balance the costs and risks of treating a man who will not get heart disease against the costs of failing to treat a man who will. This is a difficult calculation but at least we now have an understanding of the properties of the test. An alternative and popular way to display the same information is the *receiver operating characteristic* (ROC) curve which plots the sensititivity (true positive rate) against one minus the specificity (false positive rate) as seen in the second panel of Figure 2.10.

```
plot(1-Specificity,Sensitivity,type="l")
abline(0,1,lty=2)
```

A useless test (one that decides at random) would fall on the $y = x$ line whereas a very good test has a curve that is pulled out into the upper left corner. The area under the ROC curve can be used as a measure of the performance of the test and might be used to compare different tests.

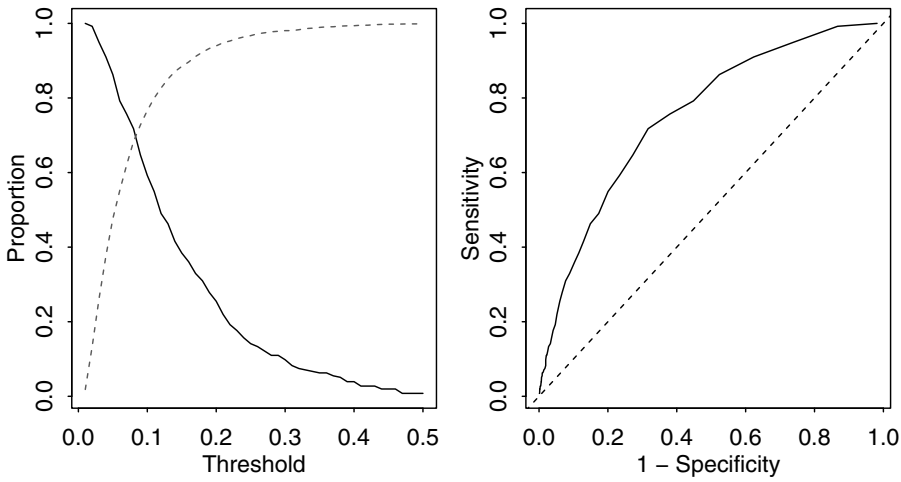


Figure 2.10 *Sensitivity and specificity for the heart disease model plotted as a function of the probability threshold on the left and as the receiver operating characteristic curve on the right.*

The proportion of variance explained or R^2 is a popular measure of fit for normal linear models. We might consider applying the same concept to binomial regression models by using the proportion of deviance explained. However, a better statistic is

due to Nagelkerke (1991):

$$R^2 = \frac{1-(\hat{L}_0/\hat{L})^{2/n}}{1-\hat{L}_o^{2/n}} = \frac{1-\exp((D-D_{null})/n)}{1-\exp(-D_{null}/n)}$$

where n is the number of binary observations and $\hat{L}_0$ is the maximized likelihood under the null. The numerator can be seen as a ratio of the relative likelihood with the $1/n$ power having the effect of a geometric mean on the observations. The denominator simply normalizes so that $0 \leq R^2 \leq 1$. For example, for the current model, the R^2 is:

```
lmodr <- glm(chd ~ age + height  +bmi + sdp + chol + dibep + cigs +
    ↪ arcus, family=binomial, wcgs)
(1-exp((lmodr$dev-lmodr$null)/3140))/(1-exp(-lmodr$null/3140))
```

```
[1] 0.14315
```

This gives the impression of a fairly poor fit when judged from the experience of linear models. However, this is misleading. In a standard linear model, it is possible for the observed and fitted values to be very close showing a strong fit and an R^2 close to one. This simply isn't possible for binary response models given the natural variation. It is quite common to see low values of Naglekerke's and other R^2 substitutes even when the model is good. For this reason, it may be best to avoid this statistic except perhaps for the purpose of comparing compatible models.

2.7 Estimation Problems

Estimation of the logistic regression model using the Fisher scoring algorithm, described in Section 8.2, is usually fast. However, difficulties can sometimes arise. When convergence fails, it is sometimes due to a problem exhibited by the following dataset. We take a subset of the famous Fisher Iris data to consider only two of the three species of Iris and use only two of the potential predictors:

```
library(dplyr)
irisr <- filter(iris, Species != "virginica") %>%  select(Sepal.Width,
    ↪  Sepal.Length,Species)
```

We plot the data using a different shape of plotting symbol for the two species:

```
(p <- ggplot(irisr, aes(x=Sepal.Width, y=Sepal.Length, shape=Species))
    ↪ +geom_point())
```

We now fit a logistic regression model to see if the species can be predicted from the two sepal dimensions.

```
lmod <- glm(Species ~ Sepal.Width + Sepal.Length, family=binomial,
    ↪ irisr)
```

```
Warning messages:
1: glm.fit: algorithm did not converge
2: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

We see that there were problems with the convergence. A look at the summary reveals further evidence:

```
sumary(lmod)
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)      -361     195973       0        1
Sepal.Width      -110      55362       0        1
Sepal.Length      132      64577       0        1
```

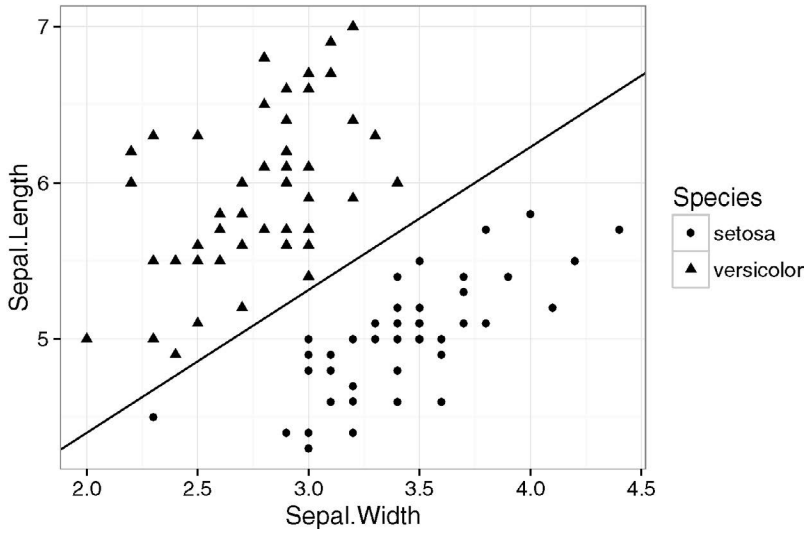


Figure 2.11 *Two species of iris by sepal length and width. The line is computed from the bias-reduced GLM fit.*

```
n = 100 p = 3
Deviance = 0.000 Null Deviance = 138.629 (Difference = 138.629)
```

Notice that the residual deviance is zero indicating a perfect fit and yet none of the predictors are significant due to the high standard errors. A look at the data reveals the reason for this. We see that the two groups are *linearly separable* so that a perfect fit is possible. We suffer from an embarrassment of riches in this example — we can fit the data perfectly. Unfortunately, this results in unstable estimates of the parameters and their standard errors and would (probably falsely) suggest that perfect predictions can be made. An alternative fitting approach might be considered in such cases called *exact logistic regression*. See Cox (1970) or Mehta and Patel (1995). Implementations can be found in the `elrm` and `logistiX` packages in R.

An alternative to exact methods is the bias reduction method of Firth (1993) and implemented in the `brglm` package of Kosmidis (2013). For the maximum likelihood estimate (MLE), $E\hat{\beta} \neq \beta$ and indeed a sensible unbiased estimator would be difficult to obtain. Firth's method removes the $O(1/n)$ term from the asymptotic bias of estimated coefficients. These estimates have the advantage of always being finite:

```
library(brglm)
bmod <- brglm(Species ~ Sepal.Width + Sepal.Length, family=binomial,
    ↪ irisr)
summary(bmod)
```

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)    -24.51      12.49   -1.96   0.0498
Sepal.Width     -8.90       2.75   -3.24   0.0012
Sepal.Length     9.73       3.33    2.92   0.0035
```

```
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 130.638  on 99  degrees of freedom
Residual deviance:   3.323  on 97  degrees of freedom
```

We can see that this results in significant predictors which we expect given Figure 2.11. We compute and display the line corresponding to a predicted probability of 1/2.

```
p + geom_abline(intercept=(0.5+24.51)/9.73, slope=8.9/9.73)
```

Instability in parameter estimation will also occur in datasets that approach linear separability. Care will also be needed in such cases.

Further Reading: See books by Collett (2003), Hosmer and Lemeshow (2013), Cox (1970), Harrell (2001), Menard (2002), Christensen (1997), Kleinbaum and Klein (2002) and Hilbe (2009).

Exercises

1. The dataset `wbca` comes from a study of breast cancer in Wisconsin. There are 681 cases of potentially cancerous tumors of which 238 are actually malignant. Determining whether a tumor is really malignant is traditionally determined by an invasive surgical procedure. The purpose of this study was to determine whether a new procedure called fine needle aspiration, which draws only a small sample of tissue, could be effective in determining tumor status.

   (a) Plot the relationship between the classification and `BNucl`.

      i. Explain why
      ```
      plot(Class ~ BNucl, wbca)
      ```
      does not work well.

      ii. Create a factor version of the response and produce a version of the first panel of Figure 2.1. Comment on the shape of the boxplots.

      iii. Produce a version of the second panel of Figure 2.1. What does this plot say about the distribution?

      iv. Produce a version of the interleaved histogram shown in Figure 2.2 and comment on the distribution.

   (b) Produce a version of Figure 2.3 for the predictors `BNucl` and `Thick`. Produce an alternative version with only one panel but where the two types are plotted differently. Compare the two plots and describe what they say about the ability to distinguish the two types using these two predictors.

   (c) Fit a binary regression with `Class` as the response and the other nine variables as predictors. Report the residual deviance and associated degrees of freedom. Can this information be used to determine if this model fits the data? Explain.

   (d) Use AIC as the criterion to determine the best subset of variables. (Use the `step` function.)

   (e) Suppose that a cancer is classified as benign if $p > 0.5$ and malignant if $p < 0.5$. Compute the number of errors of both types that will be made if this method is applied to the current data with the reduced model.

(f) Suppose we change the cutoff to 0.9 so that $p < 0.9$ is classified as malignant and $p > 0.9$ as benign. Compute the number of errors in this case.

(g) Produce an ROC plot and comment on effectiveness of the new diagnostic test.

(h) It is usually misleading to use the same data to fit a model and test its predictive ability. To investigate this, split the data into two parts — assign every third observation to a test set and the remaining two thirds of the data to a training set. Use the training set to determine the model and the test set to assess its predictive performance. Compare the outcome to the previously obtained results.

2. The National Institute of Diabetes and Digestive and Kidney Diseases conducted a study on 768 adult female Pima Indians living near Phoenix. The purpose of the study was to investigate factors related to diabetes. The data may be found in the the dataset `pima`.

(a) Create a factor version of the test results and use this to produce an interleaved histogram to show how the distribution of insulin differs between those testing positive and negative. Do you notice anything unbelievable about the plot?

(b) Replace the zero values of insulin with the missing value code `NA`. Recreate the interleaved histogram plot and comment on the distribution.

(c) Replace the incredible zeroes in other variables with the missing value code. Fit a model with the result of the diabetes test as the response and all the other variables as predictors. How many observations were used in the model fitting? Why is this less than the number of observations in the data frame.

(d) Refit the model but now without the `insulin` and `triceps` predictors. How many observations were used in fitting this model? Devise a test to compare this model with that in the previous question.

(e) Use AIC to select a model. You will need to take account of the missing values. Which predictors are selected? How many cases are used in your selected model?

(f) Create a variable that indicates whether the case contains a missing value. Use this variable as a predictor of the test result. Is missingness associated with the test result? Refit the selected model, but now using as much of the data as reasonable. Explain why it is appropriate to do this.

(g) Using the last fitted model of the previous question, what is the difference in the odds of testing positive for diabetes for a woman with a BMI at the first quartile compared with a woman at the third quartile, assuming that all other factors are held constant? Give a confidence interval for this difference.

(h) Do women who test positive have higher diastolic blood pressures? Is the diastolic blood pressure significant in the regression model? Explain the distinction between the two questions and discuss why the answers are only apparently contradictory.

3. A study was conducted on children who had corrective spinal surgery. We are interested in factors that might result in kyphosis (a kind of deformation) after surgery. The data can be loaded by

```
data(kyphosis,package="rpart")
```

Consult the help page on the data for further details.

(a) Make plots of the response as it relates to each of the three predictors. You may find a jittered scatterplot more effective than the interleaved histogram for a dataset of this size. Comment on how the predictors appear to be related to the response.

(b) Fit a GLM with the kyphosis indicator as the response and the other three variables as predictors. Plot the deviance residuals against the fitted values. What can be concluded from this plot?

(c) Produce a binned residual plot as described in the text. You will need to select an appropriate amount of binning. Comment on the plot.

(d) Plot the residuals against the `Start` predictor, using binning as appropriate. Comment on the plot.

(e) Produce a normal QQ plot for the residuals. Interpret the plot.

(f) Make a plot of the leverages. Interpret the plot.

(g) Check the goodness of fit for this model. Create a plot like Figure 2.9. Compute the Hosmer-Lemeshow statistic and associated p-value. What do you conclude?

(h) Use the model to classify the subjects into predicted outcomes using a 0.5 cutoff. Produce cross-tabulation of these predicted outcomes with the actual outcomes. When kyphosis is actually present, what is the probability that this model would predict a present outcome? What is the name for this characteristic of the test?

4. Treatment of prostate cancer depends on whether the cancer has spread to the surrounding lymph nodes. This can be determined using a surgical procedure but it would be better if noninvasive methods could be used. Load in the data and learn about the variables by:

```
data(nodal, package="boot")
help(nodal, package="boot")
```

(a) A plot consisting of a binary image of the data can be constructed as:

```
nodal$m <- NULL
image(as.matrix(nodal))
```

Improve this plot by ordering the cases on the response and labeling the axes informatively using the `axis` command.

(b) Fit an appropriate model with nodal outcome as the response and the other five variables as predictors. Is there evidence that at least some of the five predictors are related to the response?

(c) Fit a smaller model that removes `aged` and `grade` from the model. Can this smaller model be used in preference to the larger model?

(d) How much does having a serious x-ray result increase the odds of nodal involvement compared to a nonserious result? (Use the smaller model.) Give a 95% confidence interval for the odds.

(e) Fit a model with all five predictors and all their two-way interactions. Explain why the standard errors of the coefficients are so large.

(f) Use the bias-reduced model fitting method described in the chapter to fit the model of the previous question. Which interaction is largest?

(g) If the predicted response probability exceeds 0.5, the case is classified positively and, if not, negatively. Use the bias-reduced model to classify the cases in the dataset. Compare these to the actual classifications. How many were wrongly classified? Repeat this comparison for the model in (b). Do you think these misclassification rates are a reasonable estimate of how these models will perform in the future?

5. A study was conducted to determine the effectiveness of a new teaching method in economics. The data may be found in the dataset `spector`. Write a report on how well the new method works. You should include suitable graphical depictions of the data, diagnostics on your chosen model and an interpretation of the effects of the predictors on the response.

6. Additional datasets with binary responses are readily accessible for more practice:
   - `lizards` in the `brglm` package
   - `shuttle, crabs, birthwt, biopsy, bacteria` in the `MASS` package
   - `urine` in the `boot` package